

EXPERIMENT - 2

AIM - Convert the ER diagram into a relational schema.
Write complete MySQL, CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violation.

RELATIONAL SCHEMA.

Customer (

Customer ID, PK,

Name,

Email UNIQUE,

Phone No,

DOB

)

Seller (

Seller ID PK,

Seller Name,

Phone No UNIQUE

)

Address (

Address ID. PK,

Customer ID PK,

Pincode,

State,

City,

)

~~Category (~~

~~Category ID PK,~~

~~Category Name UNIQUE,~~

~~Description~~

~~)~~



Product (

Product ID PK,
Category ID FK,
Price,
Image

)

Fashion (

Product ID PK
FK

)

Orders (

Order ID PK,
Customer ID FK,
Order Date,
Order Status

)

Payment (

Payment ID PK,
Order ID FK. UNIQUE,
Payment Date,
Amount,
Payment Method,
Status

)

Electronic (

Product PK,
FK

)

Books (

Product ID PK,
FK,

Seller ID FK

)

Order-item (

Order ID. PK/FK,
item - Seq - no PK,
Product ID FK,
Quantity,
Unit - Price

)

~~Delivery (~~

~~Delivery ID. PK,
Order ID FK UNIQUE,
Courier Name,
Status,
Delivery Date~~

~~)~~



SOURCE CODE -

```
CREATE DATABASE Ecommerce DB;
```

```
USE Ecommerce DB;
```

```
CREATE TABLE Customer (
```

```
Customer ID INT Primary Key Auto-Increment,
```

```
Name Varchar (100) NOT NULL,
```

```
Email Varchar (100) NOT NULL UNIQUE,
```

```
Phone NO Varchar (15) NOT NULL,
```

```
DOB DATE NOT NULL
```

```
);
```

```
CREATE TABLE Address (
```

```
Address ID INT Primary Key Auto-Increment,
```

```
Customer ID INT NOT NULL;
```

```
Pincode Varchar (10) NOT NULL,
```

```
State Varchar (50) NOT NULL,
```

```
City Varchar (50) NOT NULL,
```

```
FOREIGN KEY (Customer ID) references Customer
```

```
(Customer ID) ON DELETE CASCADE
```

```
);
```

```
CREATE TABLE Category (
```

```
Category ID INT Primary Key, Auto-Increment,
```

```
Category Name Varchar (100) NOT NULL UNIQUE,
```

```
Description Varchar (255)
```

```
);
```



CREATE TABLE Seller (

Seller ID INT Primary Key auto-increment,
Seller Name Varchar(100) NOT NULL,
Phone No Varchar(15) NOT NULL UNIQUE.

);

CREATE TABLE Product (

Product INT Primary Key auto-increment
Category ID INT NOT NULL,
Price Decimal(10,2) NOT NULL,
Image Varchar(255),
Foreign Key (Category ID) references Category
(Category ID) ON DELETE CASCADE

);

CREATE TABLE Order (

Order ID INT Primary Key auto-increment
Customer ID INT NOT NULL,
Order Date DATE NOT NULL,
Order Status Varchar(30) NOT NULL,
Foreign Key (Customer ID) references Customer
(Customer ID) ON DELETE CASCADE

);



CREATE TABLE OrderItem (

OrderID INT NOT NULL,

item-seq-no INT NOT NULL,

ProductID INT NOT NULL,

Quantity INT NOT NULL,

Unit-Buy DECIMAL (10,2) NOT NULL,

Primary Key (OrderID, item-seq-no)

Foreign Key (OrderID) references orders (OrderID)
ON DELETE CASCADE

Foreign Key (ProductID) references Product (ProductID)
ON DELETE CASCADE

);

CREATE TABLE Payment (

PaymentID INT Primary Key ~~OrderID - amount~~

OrderID INT NOT NULL UNIQUE,

Payment Date NOT NULL,

Amount decimal (10,2) NOT NULL,

Payment Method Varchar (30) NOT NULL,

Status varchar (30) NOT NULL,

Foreign Key (OrderID) references orders (OrderID)
ON DELETE CASCADE

);



CREATE TABLE Delivery (

Delivery ID INT Primary Key - auto-increment

Order ID INT NOT NULL UNIQUE,

Carrier Name VARCHAR (100) NOT NULL,

Status VARCHAR (30) NOT NULL,

Delivery Date DATE,

Foreign Key (Order ID) references Order
(Order ID) on DELETE CASCADE

);

Inserting Sample Data -

Insert into Customer (Name, Email, Phn No, DOB) VALUES

("Aryan", "aryan@gmail.com", "8799258161", "2008-05-01"),

("Naman", "naman@gmail.com", "7379217290", "2006-09-14"),

Insert into Category (Category Name, Description) VALUES

("Electronics", "Electronic Products"),

("Fashion", "Fashion Products"), ("Book", "Books")

Insert into Seller (Seller Name, Phn No) VALUES

("Tech Store", "9123456789"),

("Book Store", "9234567891");



Insert into Product (Category ID, Price, Image) VALUES

(1, 50000, "dokter.jpg"), (2, 999, "Shirt.png"),
(3, 999, "door.png");

Insert into Electronic Values (1);

Insert into Fashion Values (2);

Insert into Books (3, 2);

Insert into Address (Customer ID, Province, State, City) VALUES

(1, "20800", "Utara Pradesh", "Kamphur"),
(2, "39500", "Gugiat", "Surat");

Insert into Orders (Customer ID, Order Date, Order Status) VALUES

(1, "2026-08-19", "Placed"),
(2, "2026-08-19", "Shipped");

Insert into OrderItem VALUES

(1, 1, 1, 50000), (2, 1, 2, 999);

Insert into Payment (Order ID, Payment Date, Amount, Payment Method,
Status) VALUES

(1, "2026-08-19", 50000, "UPI", "SUCCESS"),
(2, "2026-08-19", 1998, "Card", "SUCCESS");



Insert into Delivery (OrderID, Courier Name, Status, DeliveryDate)
VALUES

(1, "Delivery", "Delivered", "2026-08-20"),
(2, "DTDC", "Pending", NULL);

Referential Integrity Violations -

Insert into orders (CustomerID, OrderDate, OrderStatus)
VALUES

(999, "2026-08-19", "Placed");

OUTPUT:-

ERROR: Cannot add or update a child row:
A foreign key constraint fails

Unique Constraint Violation -

Insert into Customer (Name, Email, Phone No, DOB) VALUES
("Test", "aryam@gmail.com", "9999999999", "2025-01-01");

Output:-

ERROR: Duplicate entry "aryam@gmail.com"
for key 'customer.Email'



Select * from Customer ;
Select * from Product ;
Select * from Orders ;
Select * from OrderItem ;

ON DELETE SET NULL

DELETE From Seller Where SellerID = 2;

Select * From Books;

ON DELETE CASCADE

DELETE From Customer where CustomerID = 2;

Select * From Orders where CustomerID = 2;

OUTPUT -

CustomerID	Name	Email	Pham No.	DOB
1	Aryam	aryam@gmail.com	8749256161	2008-05-01
2	Narman	narman@gmail.com	7379217290	2006-09-14



Product ID	Category ID	Price	Group
1	1	50000.00	Jaketah .img
2	2	999.06	Shurt .img
3	3	599.00	Book .img

Order ID	Customer ID	Order Date	Order Status
1	1	2026-08-19	Placed
2	2	2026-08-19	Shipped

Order ID	Item - Ser - No	Product ID	Quantity	Unit - Price
1	1	1	1	50000.06
2	1	2	2	999.06

Handwritten signature

Product ID	Seller ID
3	NULL

Empty Set

